

Citadel 2024 Spring Invitational Datathon

Shades of Green & Gray: Exploring the Socioeconomic Impacts of Energy Transition in the United States

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1 Executive Summary

The transition from fossil fuels to renewable energy sources is an issue of critical importance for the United States: we cannot ignore the acceleration of climate change, but at the same time, the US is extremely dependent on the fossil fuel industry.

This report examines the complex relationship between energy-related variables and key socioeconomic indicators across different states. By utilizing random forest and LASSO regressions, as well as a Long Short-Term Memory (LSTM) neural network for forecasting oil prices, we uncover nuanced insights into how

energy variables influence economic and health outcomes.

Our findings reveal a significant dependence of the U.S. economy on the fossil fuel industry, with petroleum consumption and oil exports strongly influencing employment and income levels. However, there are signs of a positive impact from renewable energy consumption and production on these socioeconomic indicators in states like California and South Dakota, suggesting a gradual shift towards greener sources of wealth generation. The relationship between energy variables and healthcare expenditure proved to be less conclusive, suggesting a need for further research with more granular data.

2 Introduction

2.1 Motivation

Recall a common scene in dystopian fiction: A mega-corporation dumps industrial waste from its manufacturing plants into the natural environment around it, to the detriment and despair of powerless local communities. This has been the lived reality of people living near fracking sites in the United States. A huge corpus of literature has been published on this matter: Children in Pennsylvania living near fracking sites, for instance, have been found to be 2-3 times as vulnerable to leukemia. [3]

Health impacts are only one of the myriad side-effects of the fossil fuel industry in the United States. Studies in other regions give us a glimpse of the socioeconomic benefits that could arise from a transition to renewable energy sources. In Africa, improvements in energy access have been found to decrease poverty and increase life expectancy, and in North Macedonia, increased energy efficiency has been positively correlated with employment. [1] [2]

However, it was also found in France that rising energy prices - and thus decreased emissions - were associated with decreased employment, and an Oxford study found that there is a trade-off between climate control and income equality. [4] [5]

Perhaps it is not so simple as to say that fossil fuels are bad and renewables are good. While a transition away from oil and towards renewables is ultimately necessary for the United States, it remains to be seen which of the negative and positive side-effects seen in the aforementioned studies also apply here.

Hence, it behooves us to explore how variables such as energy consumption, commodity prices, and emissions might positively or negatively affect socioeconomic indicators in the United States, filling a gap in the literature on this subject and taking a small step towards more informed national and state-level energy policy.

2.2 Variables of Interest

On a high level, we intend to explore the effects of the following aggregate energy-related variables on the following socioeconomic indicators in the United States:

Aggregate energy-related variables	Socioeconomic indicators
Energy consumption	Monthly unemployment per state
Energy production	Quarterly personal income per state
Net energy stocks on hand	Yearly physician expenditure per state
Net emissions	
Net oil imports	
Energy prices	

We further divide each aggregate energy-related variable into sub-variables; for example, we split energy consumption into different classes of energy, like renewables, coal and natural gas. **This allows us to draw**

more specific conclusions, and importantly, to differentiate between green energy and fossil fuels. A full list of our sub-variables is available in [Appendix A](#).

2.3 Research Questions

To rigorously define the relationship between each energy-related variable and socioeconomic indicator, there are three specific questions we wish to answer:

1. How significantly does each variable influence each socioeconomic indicator?
2. Does each significant variable exhibit a positive or negative relationship with each socioeconomic indicator?
3. Also, how does this differ by state, given that different states have different energy profiles? (e.g. Texas is a major oil producer, while South Dakota is a major renewable energy producer.)

2.4 Non-Technical Summary

We were able to arrive at significant and logical predictors for unemployment and income in the US. Specifically, we found that employment and income are strongly correlated with the production and consumption of fossil fuels, indicating that **the US is still, unfortunately, heavily reliant on the fossil fuel industry.**

However, our analysis was not without points of hope: In California and South Dakota, states that consume and produce large amounts of renewable energy respectively, we found that renewable energy consumption and production were also moderately significant and positively correlated with employment and income, signalling that **the US might be undergoing a shift in the means of generating wealth, from fossil fuels to renewables.**

Our findings for healthcare were less conclusive, which we surmise was due to the comparatively low amount of data we used. However, we still found that **energy production in general is significantly and positively correlated with healthcare expenditure.**

Finally, we note that crude oil prices were a moderately significant predictor for employment. Given that we were provided rich time series data for this, we built a model to predict crude oil prices themselves, allowing us to then make predictions for future US employment.

2.5 Methodology

2.5.1 Datasets

For our energy-related variables, we found the provided datasets sufficient. Specifically, we used the 5 provided energy datasets, as well as the dataset on commodity prices.

For our socioeconomic indicators, the provided datasets were not sufficiently granular. The American Community Survey (ACS) data only provided 12 observations per state, one for each year from 2010-2021. While we used these datasets for EDA, we decided against modelling with this data since our features would outnumber our observations. Instead, we found public datasets from the [US Bureau of Economic Analysis](#) for personal income, [US Bureau of Labor Statistics](#) for unemployment, and [Centers for Medicare and Medicaid Services](#) for healthcare expenditure.

2.5.2 Cleaning

To clean and consolidate our predictor variables (i.e. the energy and commodities datasets), we performed these steps:

1. For each dataset, we removed all irrelevant columns. For instance, in the energy datasets, the 'YYYYMM' column provided the same information as the 'Date' column, so we dropped the former.

2. For each dataset, we indexed each entry by the 'Date' column, which was common to all datasets. We converted this to a 'datetime' object for ease of manipulation.
3. We converted our energy data values to be in the same unit (Btu) for ease of comparison.
4. We resampled our data to be at the appropriate frequency for each of our socioeconomic indicators (monthly, quarterly, or yearly). For our energy data, we summed the values for each month to get yearly and quarterly totals, whereas for our commodities data, we took the average price for each quarter or year.

As for our response variables (i.e. the personal income, unemployment, and healthcare datasets), after removing irrelevant columns, indexing by date and standardizing values, we created one dataframe of response variables for each state, hence creating 50 dataframes per dataset (or 51, if the dataset also included data for the US as a whole).

This approach would allow us to fit different models for each state, allowing us to investigate how the relationship between our predictor variables and response variables might differ by state.

2.5.3 Modelling

An examination of our cleaned predictor dataframe found that some variables have a moderate positive correlation.

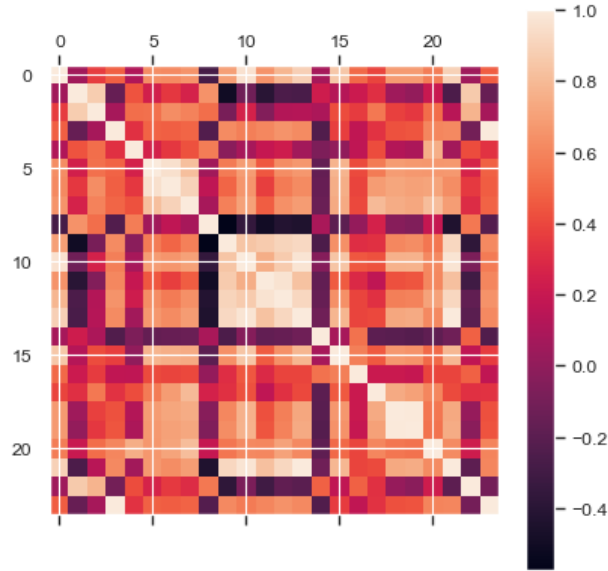


Figure 1: Correlation heatmap between predictor variables.

For each state's response data, we fit two models:

1. A random forest regression model, to determine the significance of each predictor variable on each response variable.
2. A LASSO regression model, to correct for any collinearity between predictor variables and to determine the relationship (positive or negative) between each predictor variable and each response variable.

As previously mentioned, we also set out to build a predictive model for the average daily crude oil price. To do this, we used a Long Short-Term Memory (LSTM) neural network, training it on the daily spot prices for Brent and WTI crude oil.

3 Exploratory Data Analysis

3.1 Energy Consumption

In Figure 2 we can see the total monthly energy consumption, including coal, petroleum, renewable and natural gas from 1873 to 2023 in Quadrillion BTU. We draw three main insights about energy consumption:

1. First, energy consumption has gone up between 1980 and the early 2000s but have been relatively stable on a year by year basis since approximately 2000.
2. Second, monthly energy consumption is highly seasonal. We can see this effect even more in Figure 3 which shows the average monthly energy consumption from 1973 to 2023.
3. Finally, and most importantly, Figure 5 shows that petroleum is the most consumed energy class in the US on a yearly basis, followed by natural gas, coal, and renewables.

We anticipate that this would result in petroleum-related predictors having an outsized impact compared to predictors for other energy classes.

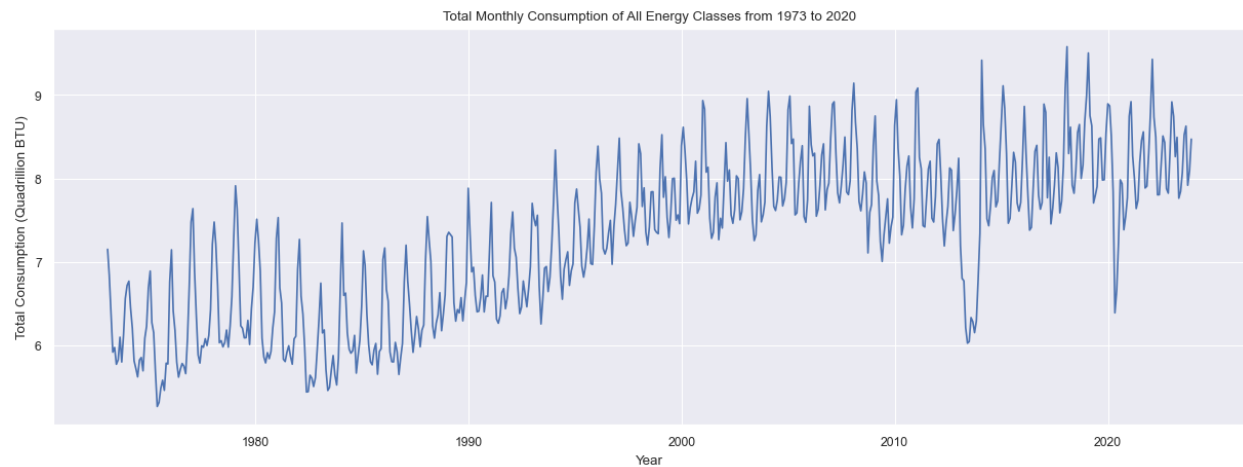


Figure 2: Total Monthly Consumption of All Energy Classes (Petroleum, Renewable, Coal, and Natural Gas from 1973 to 2023

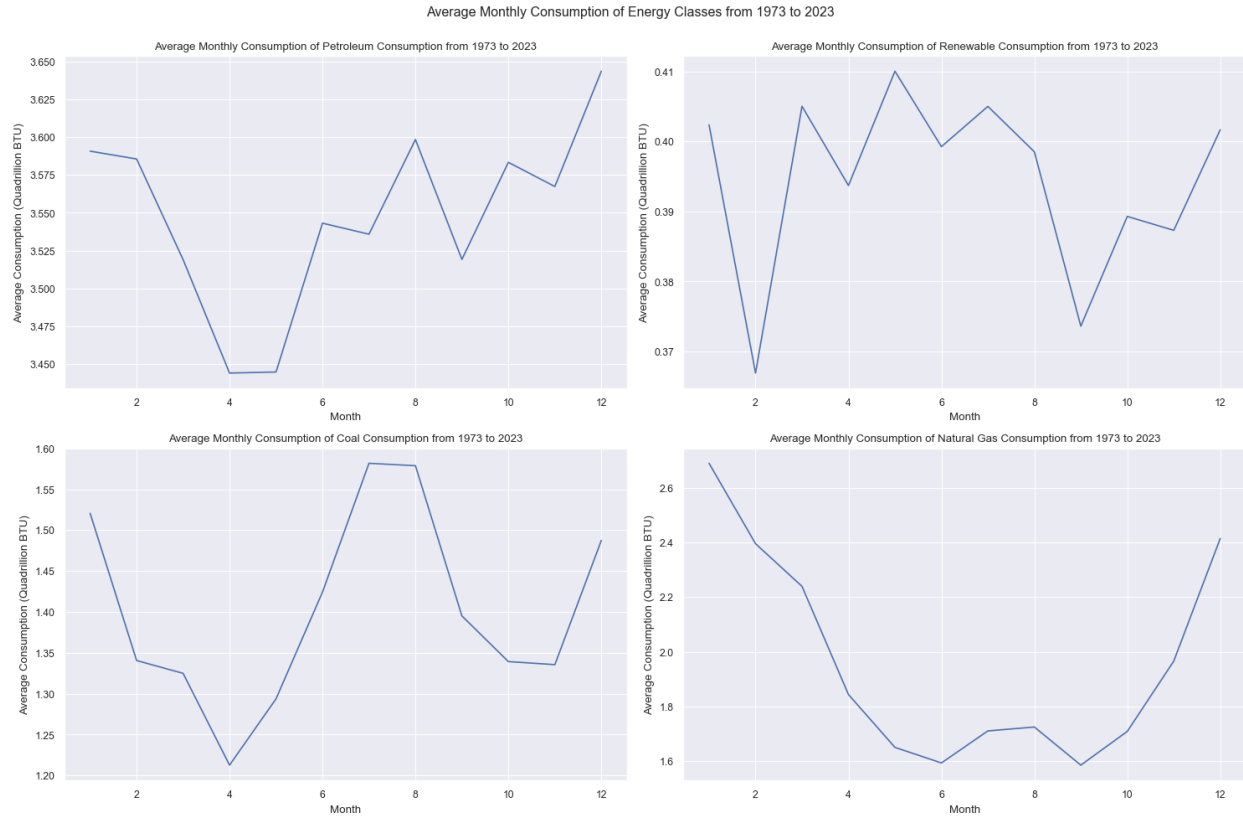


Figure 3: Average Monthly Consumption of Energy by Class from 1973 to 2023

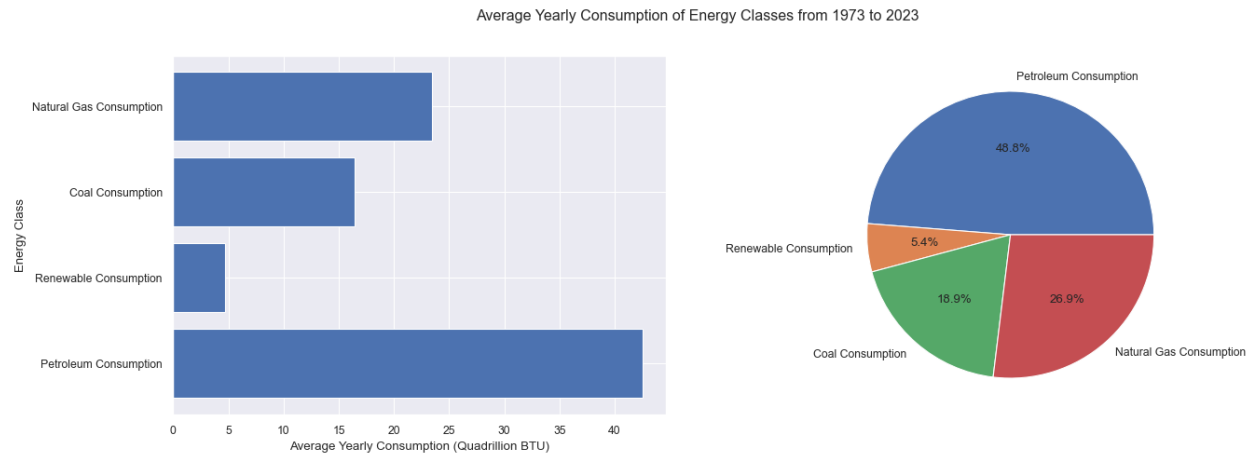


Figure 4: Average Yearly Consumption of Energy Classes from 1973 to 2023

3.2 Energy Production, Imports & Exports

We notice that there has been an aggregate uptrend in energy production in the US across all energy classes, as shown in Figure 5. This corresponds to our exploration of the US' energy imports and exports in Figure 6, which shows an increase in energy exports even while imports have remained relatively constant for the past few years (for non-OPEC countries) or even decreased (for OPEC countries).

We anticipate that this would result in production and export-related predictors having an outsized impact compared to import-related predictors.

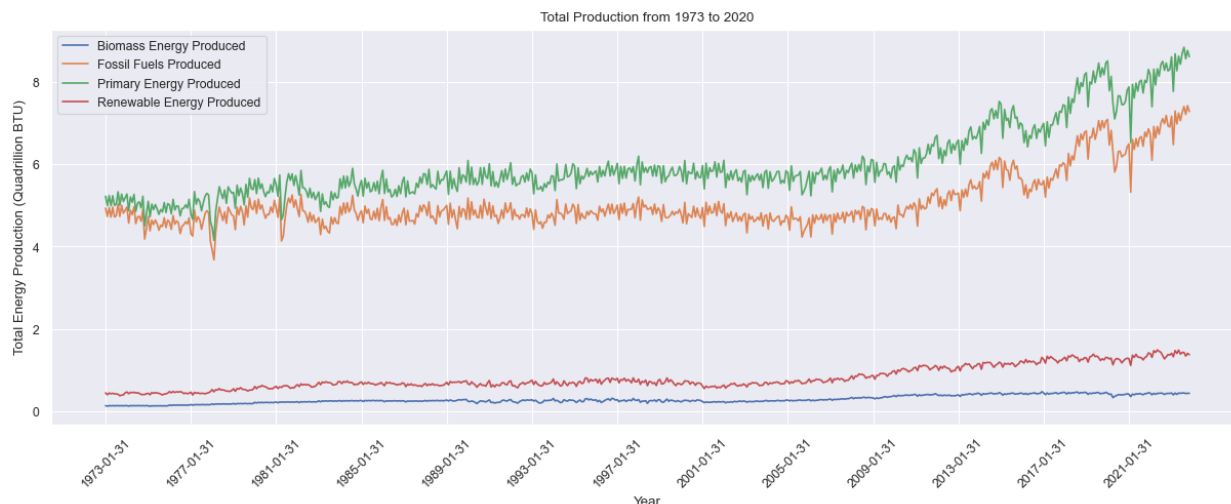


Figure 5: Average Monthly Energy Production By Energy Class in Quadrillion BTU from 1973 to 2023

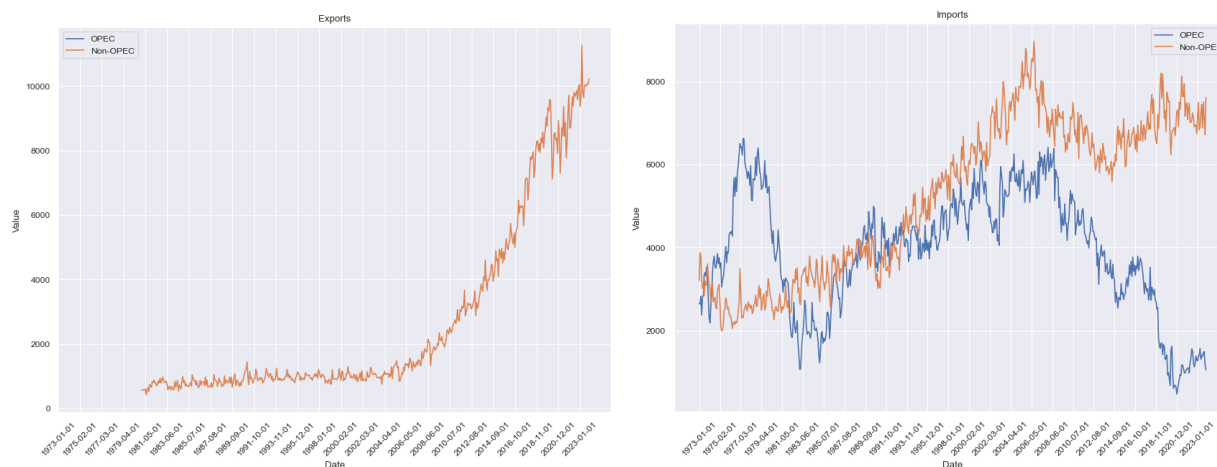


Figure 6: Average Monthly Energy Exports & Imports – note that the US only exports to non-OPEC countries

3.3 Net Energy Stocks on Hand

We find that natural gas stocks (shown in an orange color in Figure 7) a strong seasonal pattern, with peaks and troughs occurring regularly throughout the years. Upon closer inspection we find that the times of year when gas stocks are high are before winter months, possibly in anticipation of higher winter usage, while troughs represent end-of-season depletion.

The line for coal stocks (shown in blue in Figure 7) exhibits a general decline over time, particularly noticeable after the year 2000. This suggests a decreasing reliance on coal as an energy stock over the years, which could be due to a variety of factors, such as changes in energy policy, shifts to renewable energy sources, or decreased demand for coal energy.

Further, 8) shows that petroleum stocks are in general much lower and display a general upwards trend since the 1970 with a stark decline from 2020 onwards.

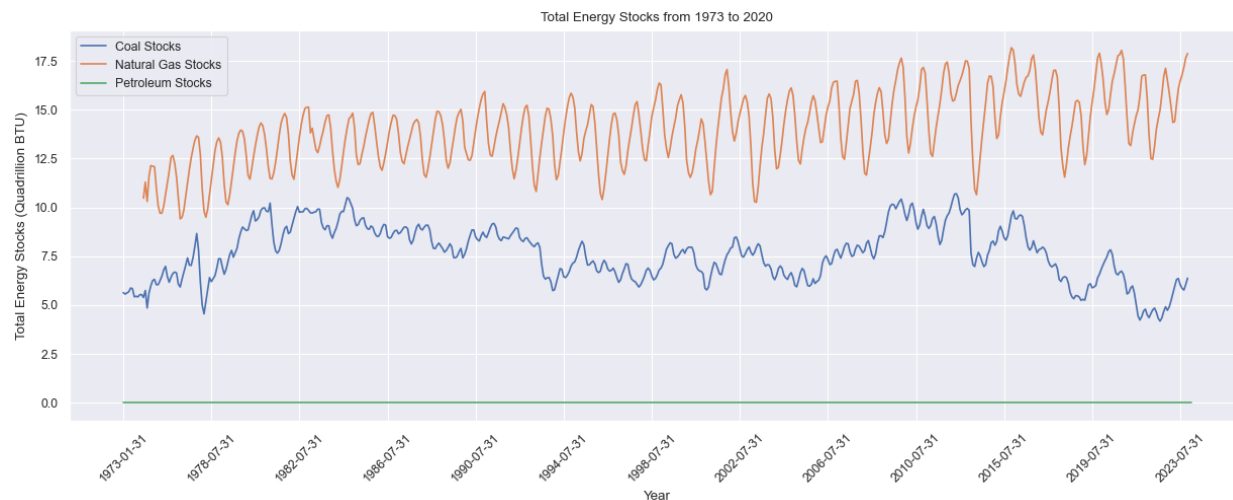


Figure 7: Monthly Energy Stocks on Hand from 1973 to 2023

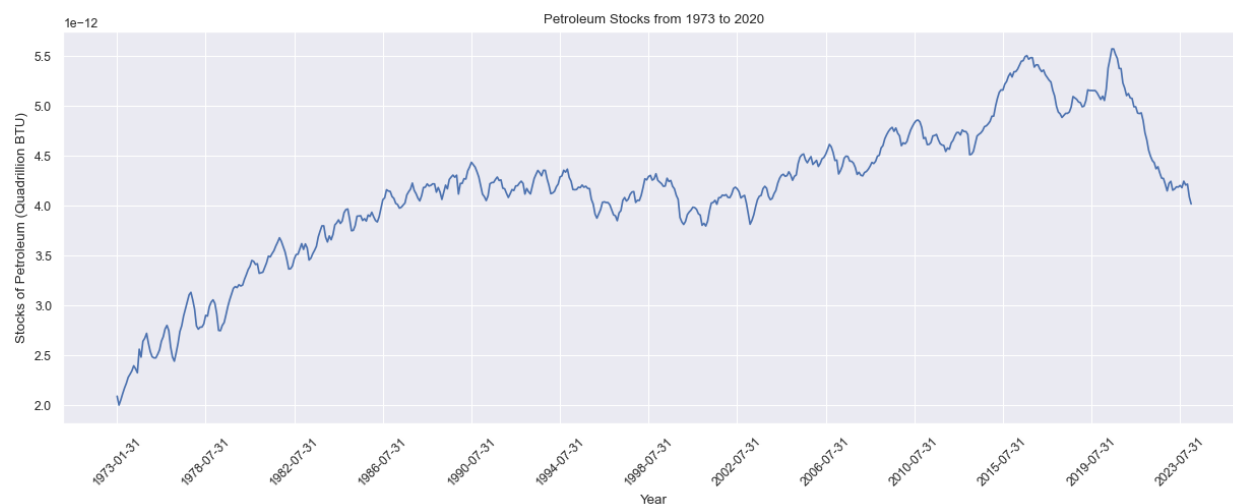


Figure 8: Monthly Petro Stocks on Hand from 1973 to 2023

3.4 Energy Prices

While natural gas prices have, by inspection, tended to revert to their mean from 1997 to 2024, we see more significant changes in Brent and WTI prices. We postulate that this could result in Brent and WTI prices having an increased influence on our chosen socioeconomic indicators, compared to natural gas.

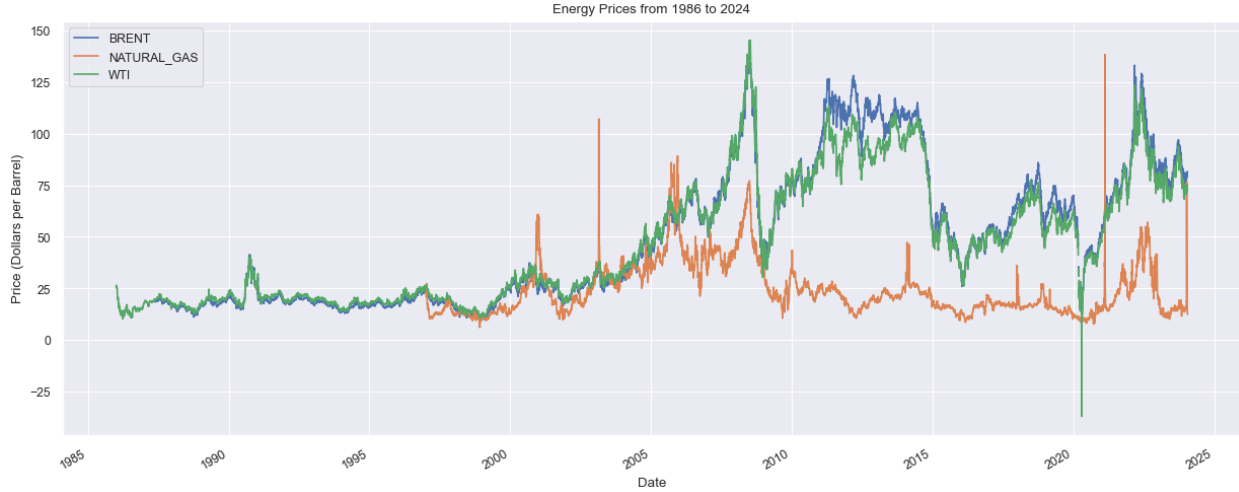


Figure 9: Energy Prices from 1986 to 2024

3.5 Monthly Unemployment per State

Figure 10 and 11 showcase the a map of the United States that is color-coded to represent the average unemployment and percentage change in unemployment rates for each state over the 12-year period respectively.

For the average unemployment rate from 2010 to 2022 in Figure 10 we observe that the average unemployment rate for the time-span lies between 3 to 9 percent. We also observe a noticeable regional pattern where the central states of the US tend to have lower average unemployment rates (bluer), while some states on the West Coast and Southern regions show higher averages (redder). For the percentage change in unemployment in Figure 11 we observe that a majority of states show a decrease in unemployment. This fits well with our intuition as we have seen a bull run in the US economy from 2010 to the beginning of the COVID-19 pandemic.

Overall, the two plots highlight the fact that there are significant disparities between statewide socioeconomic indicators, which led us to build models for each state rather than just the US as a whole.

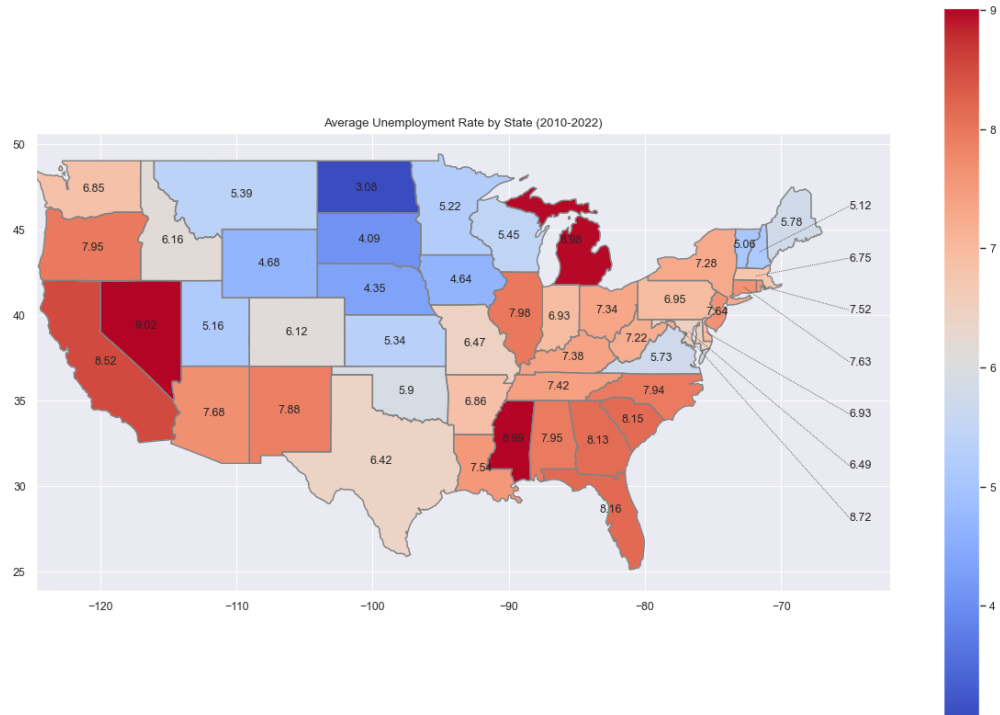


Figure 10: Average Unemployment Rate by State (2010-2022)

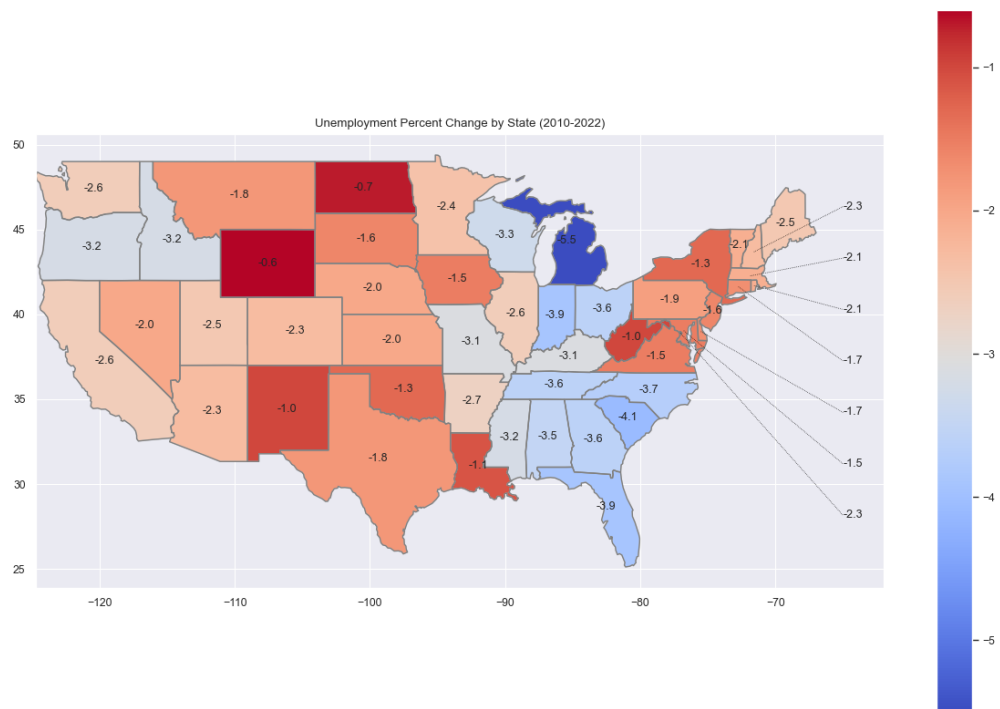


Figure 11: Unemployment Percent Change by State (2010-2022)

4 Modelling

4.1 Impact of Energy-Related Variables on Socioeconomic Indicators

Although we fit models to each state, we will focus on the following states of interest:

1. New Mexico, the top oil producer in the US after Texas.
2. California, the top energy consumer in the US after Texas.
3. South Dakota, the top green energy producer in the US.
4. The United States as a whole, if data for this was available.

We decided against using Texas because its status as both a top oil producer and energy consumer might confound our conclusions (we wouldn't know which characteristic, producer or consumer, to attribute certain conclusions to).

4.1.1 Unemployment

We first investigate the relationship between our predictors and the monthly unemployment in each state from 1976 to 2022.

In New Mexico (Figure 12), we see from our random forest regression's feature importances that **petroleum consumption in the US had by far the most significant impact on unemployment**. This is in line with our hypothesis formed during EDA.

Looking at our LASSO regression results, we see further that there is a logical **inverse relationship between unemployment and petroleum consumption**.

We also see that renewable energy consumption is positively correlated with unemployment in New Mexico, confirming that **a negative side-effect of transitioning to renewables would be unemployment in the fossil fuel industry**.

In California (Figure 13), we similarly see that **petroleum consumption has the most significant impact on unemployment, and with the same inverse relationship**. This makes sense given California's status as a top energy consumer.

Finally, in South Dakota (Figure 14), we see salient evidence of the state's commitment to green energy: **Stocks of coal had the most significant impact on unemployment, but with a strong positive relationship**. This implies that as the demand for non-renewable energy increases, people are driven out of the renewable energy industry, and vice-versa.

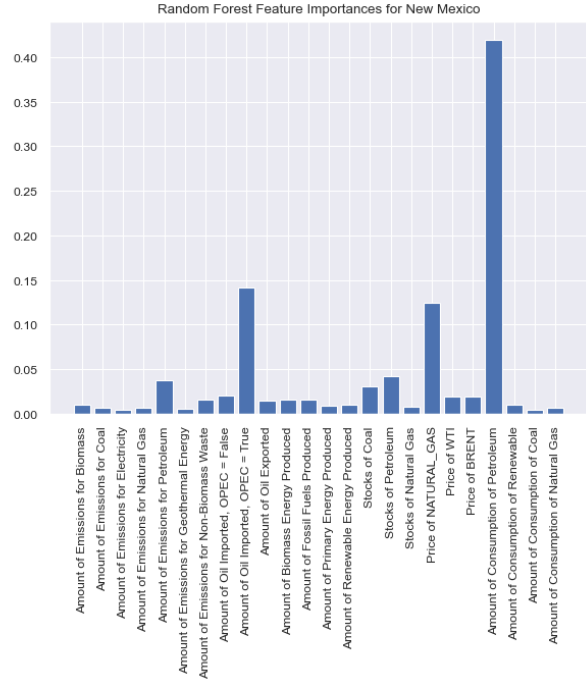


Figure 12: LASSO & Random Forest Regression Results for Unemployment in New Mexico

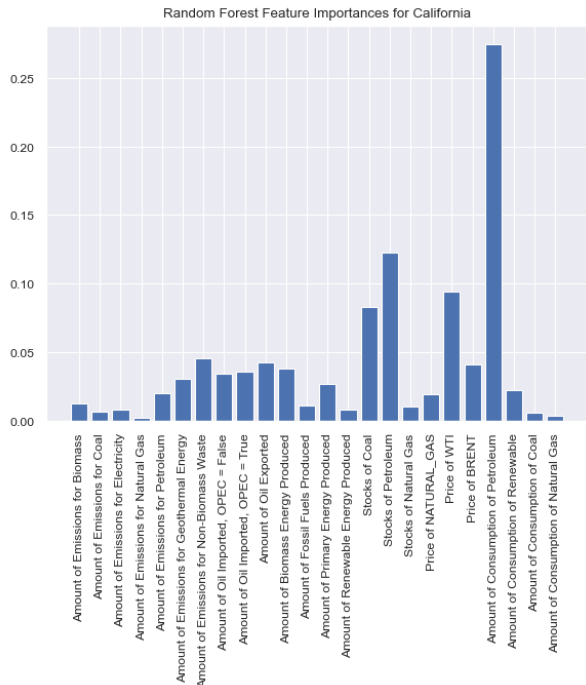
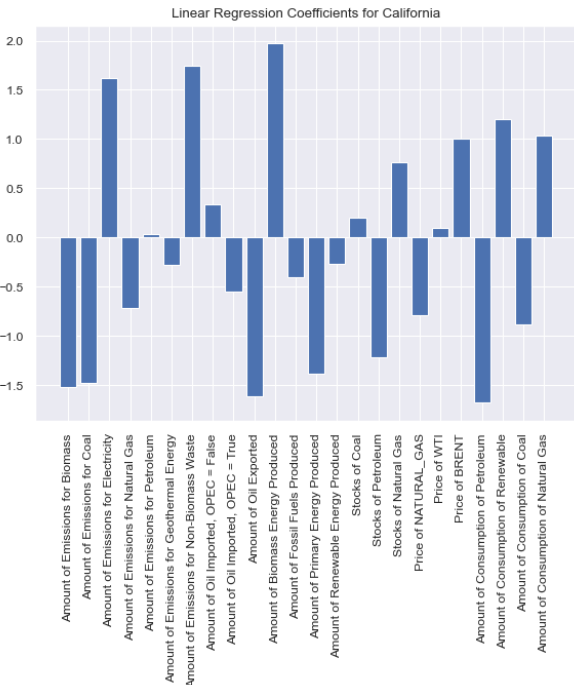


Figure 13: LASSO & Random Forest Regression Results for Unemployment in California

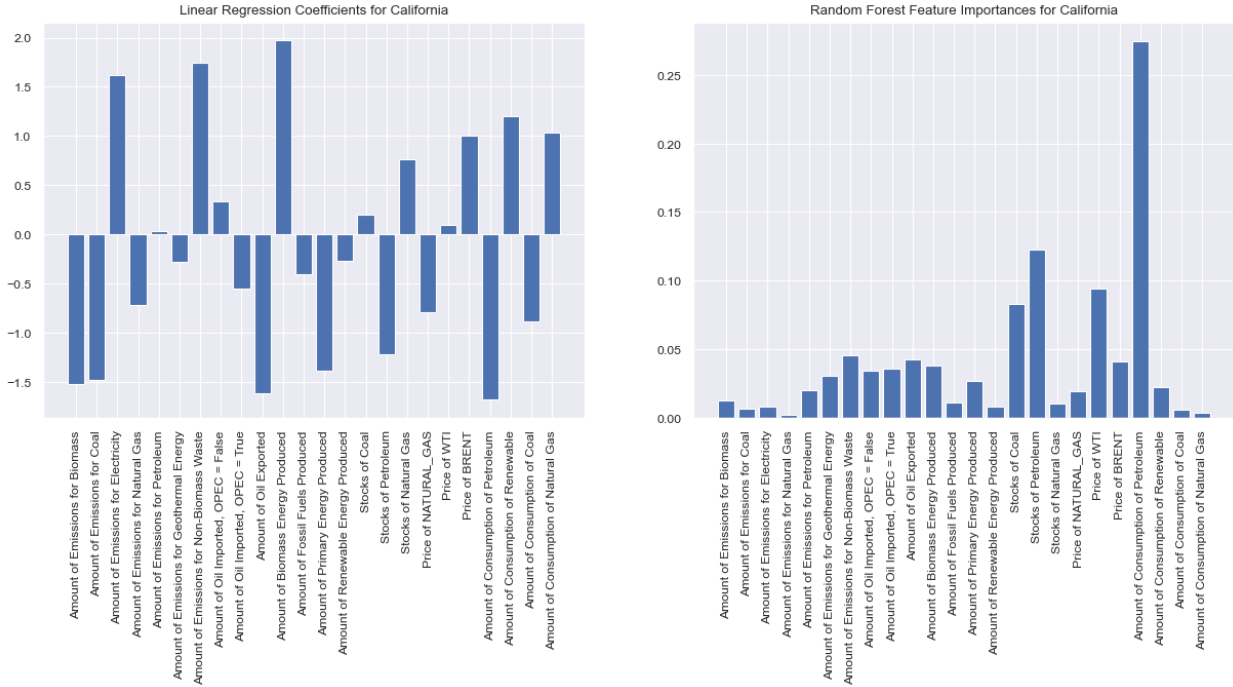


Figure 14: LASSO & Random Forest Regression Results for Unemployment in Dakota

4.1.2 Personal Income

Next, we analyzed the impact of our predictors on quarterly personal income (per capita) from 1948 to 2020.

For New Mexico and California (Figures 19 and 20, we observe that **the amount of oil exported is the most significant predictor for personal income, with a positive correlation..** For New Mexico, this is an intuitive result. For California, after further research, we realized that this makes sense because the state is still the 7th-largest crude oil producer in the US, demonstrating its reliance on oil exports to generate wealth. This is in line with our hypothesis during EDA that oil exports would be more significant predictors than imports, given the US' decreasing reliance on imported energy.

Interestingly, in New Mexico, we also see that **coal emissions are a moderately important predictor of income, but with a negative correlation.** This is a potential, albeit indirect indicator that emissions harm worker health, impacting their ability to work.

In California, we also see an encouraging result. **The consumption of renewable energy is a moderately important and positively-correlated predictor of income.** We see this same pattern reflected for the US as a whole (Figure 22), where renewables consumption is the second most important predictor for personal income other than oil exports.

In South Dakota (Figure 21), we see an outsized influence of oil exports on personal income once again. However, this is likely due to the fact that the fossil fuel industry is aggregately much larger than the renewables industry in the US as a whole. **If we just compare the variables for energy production, however, we see that renewable energy production is the most important predictor, with a positive correlation.** This, along with our findings for California, are hopeful signs of the ability for the renewables industry to generate wealth in the US.

Finally, we also see that when we just look at the feature importances for prices of natural gas, WTI and Brent, our hypothesis during EDA was confirmed that prices of WTI and Brent are more important

than natural gas when it comes to predicting unemployment. Compared to the rest of the variables, we see that WTI and Brent are also themselves moderately (approx. top 10) important predictors.

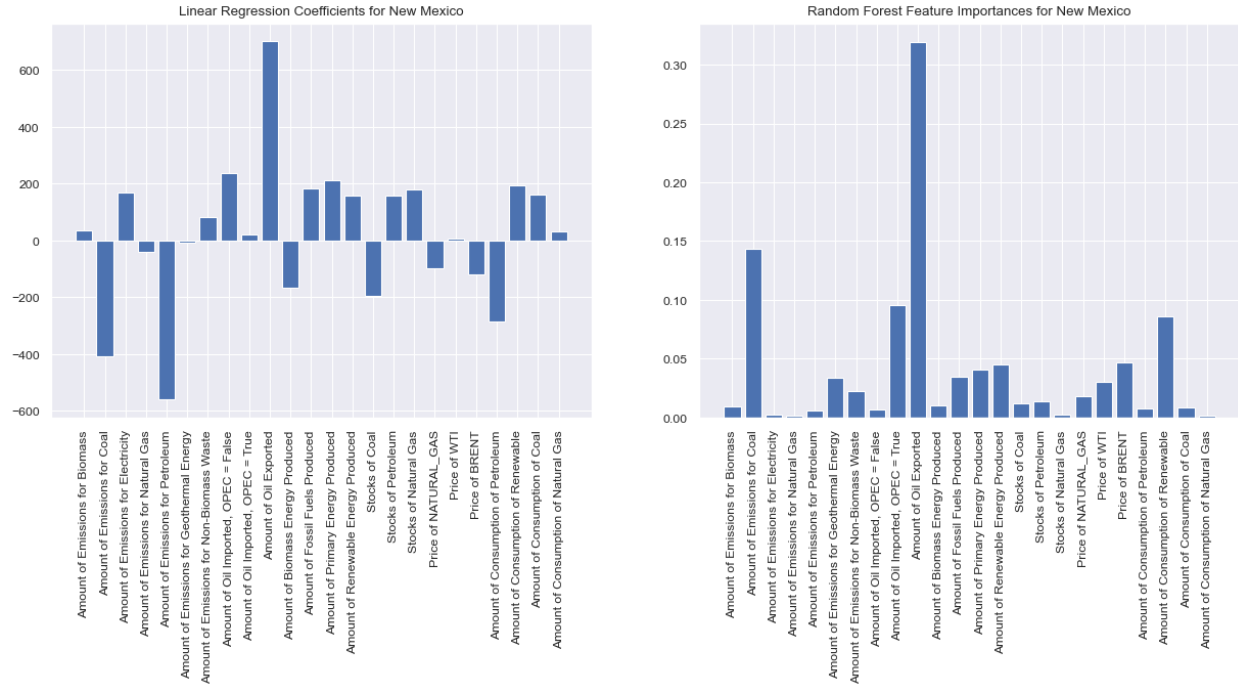


Figure 15: LASSO & Random Forest Regression Results for Personal Income in New Mexico

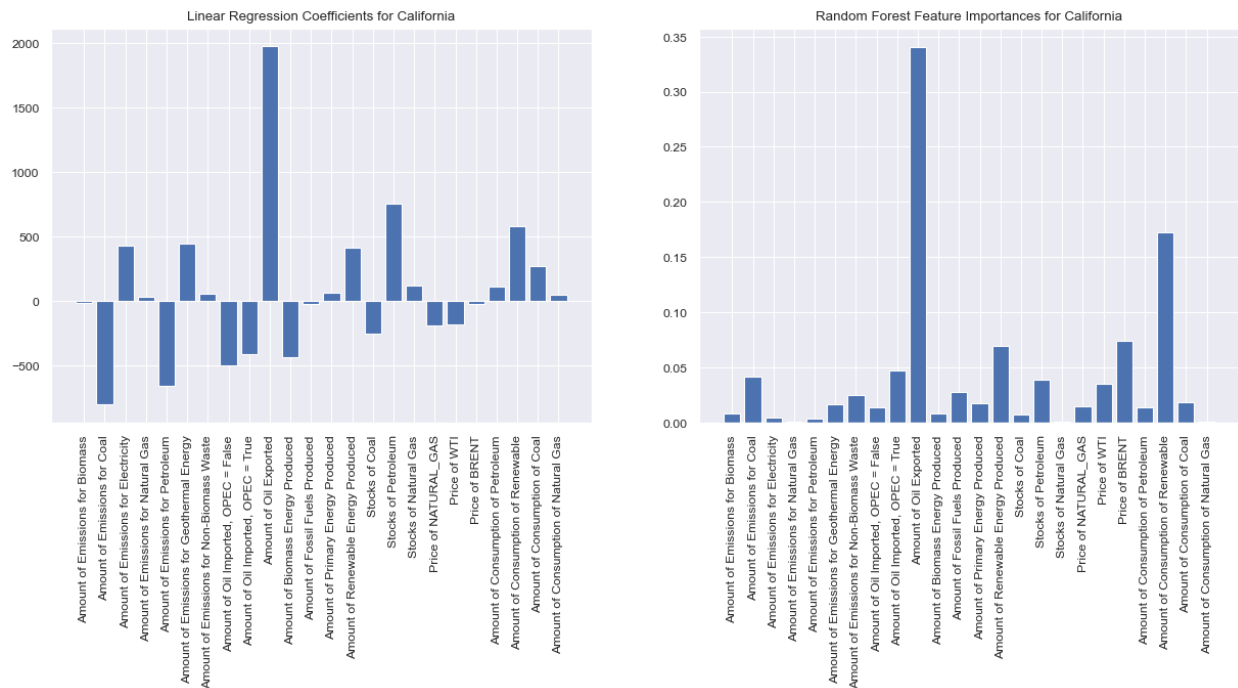


Figure 16: LASSO & Random Forest Regression Results for Personal Income in California

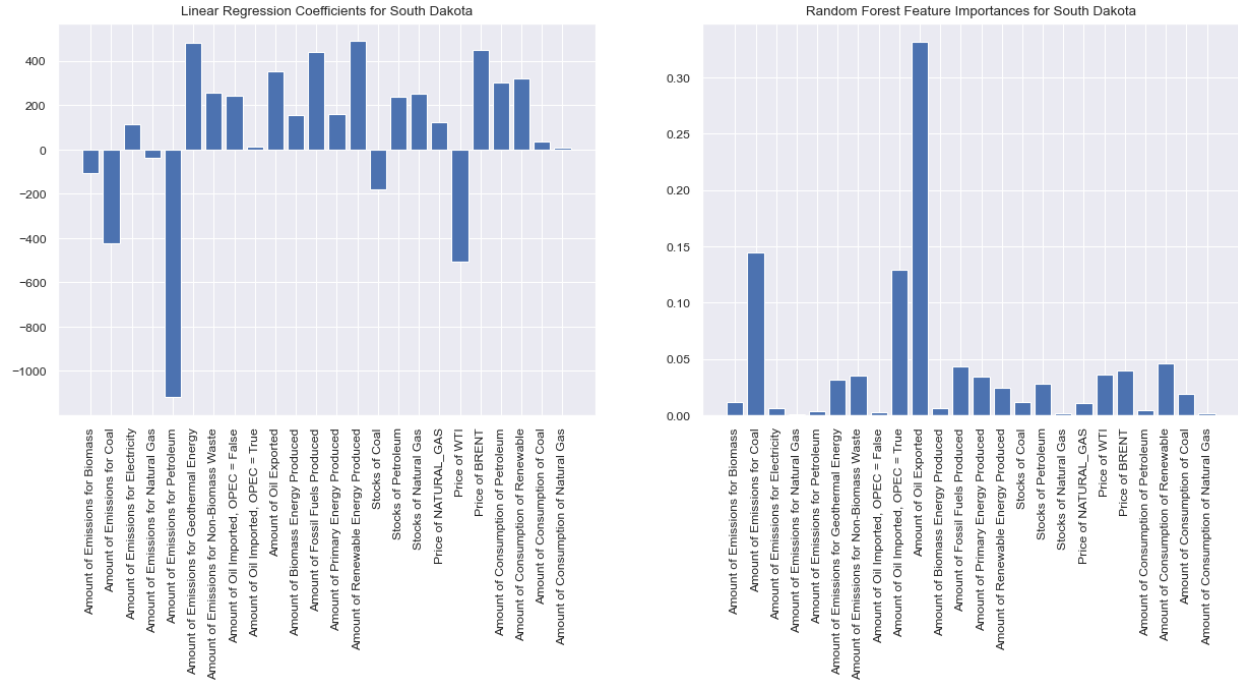


Figure 17: LASSO & Random Forest Regression Results for Personal Income in Dakota

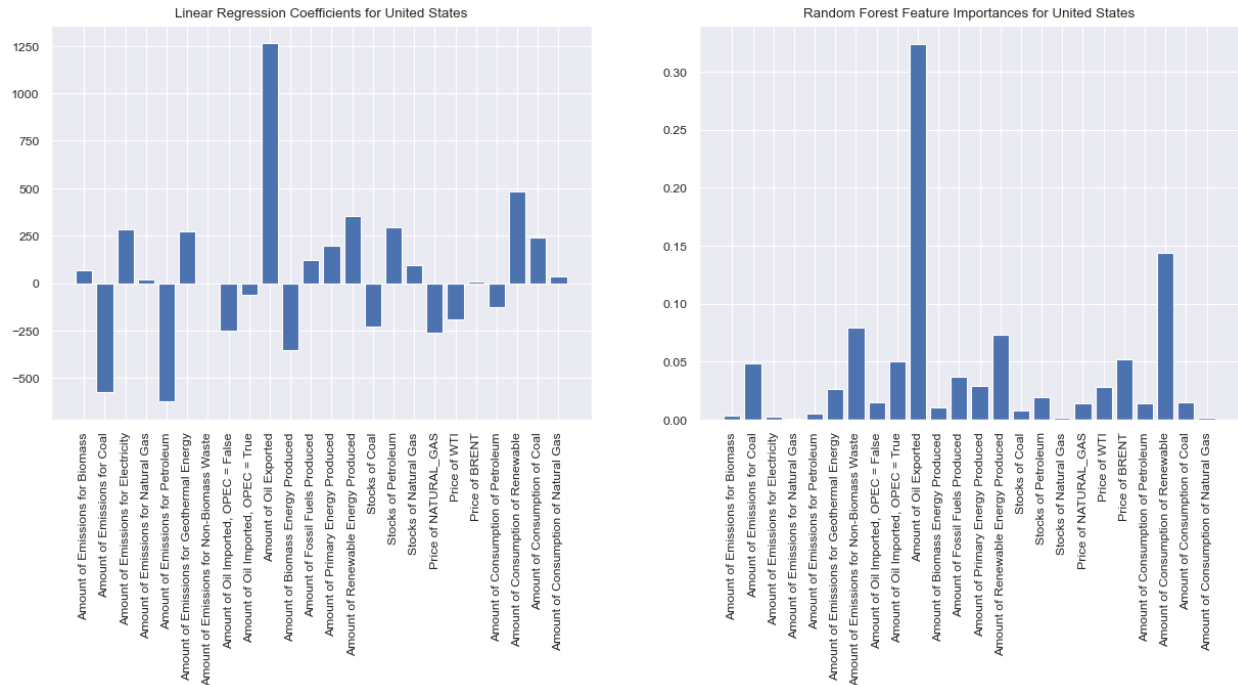


Figure 18: LASSO & Random Forest Regression Results for Personal Income in the US

4.1.3 Healthcare Expenditure

Finally, we examined the relationship between our variables and personal yearly healthcare expenditure (per capita) on physicians & clinicians from 1991 to 2020.

Unfortunately, our results for this section were less conclusive. We see that for all 3 states of interest as well as in the whole of the US, many different variables were identified as significant by our random forest regression model, with positive and negative relationships with healthcare expenditure in no clear pattern.

We believe that this is because the dataset had significantly less observations (only 20, one per year) than our other 2 datasets, impacting model precision.

We decided to fit our models to our aggregate energy-related variables instead (see the table in section 2.2), reducing the number of predictors since we have a small number of observations. Our results for each state, as well as the whole of the US, were similar; we display our results for the US in Figure 23). **We see that net energy production has the most significant impact on healthcare expenditure, and a positive relationship.**

That said, the question remains of whether increased healthcare expenditure might be due to increased rate of illness, and/or due to the increased ability to spend on healthcare, both of which could occur when more energy is produced.

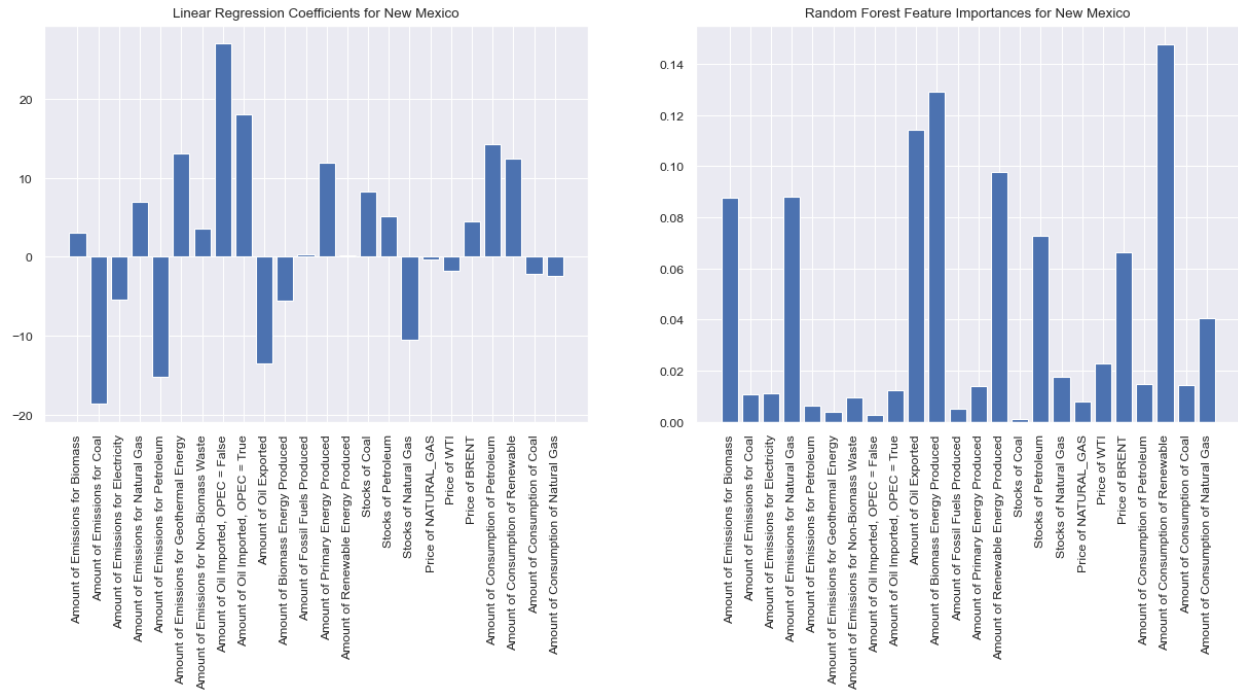


Figure 19: LASSO & Random Forest Regression Results for Physician & Clinician Expenditure in New Mexico

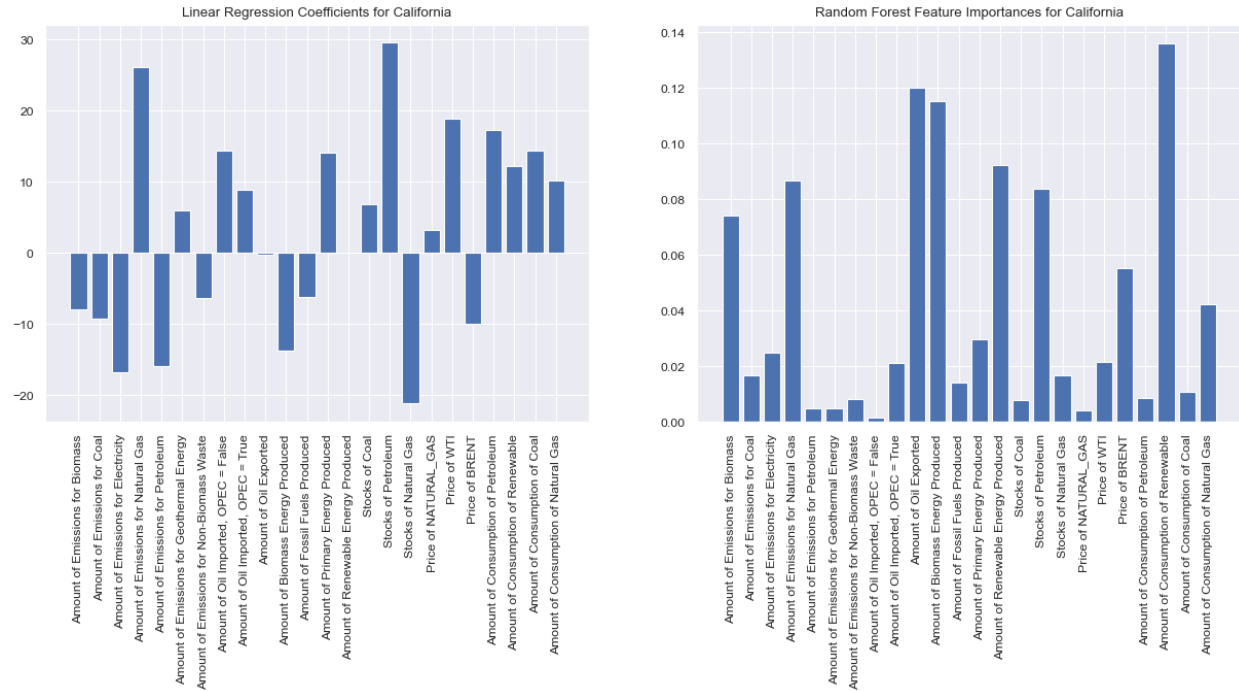


Figure 20: LASSO & Random Forest Regression Results for Physician & Clinician Expenditure in California

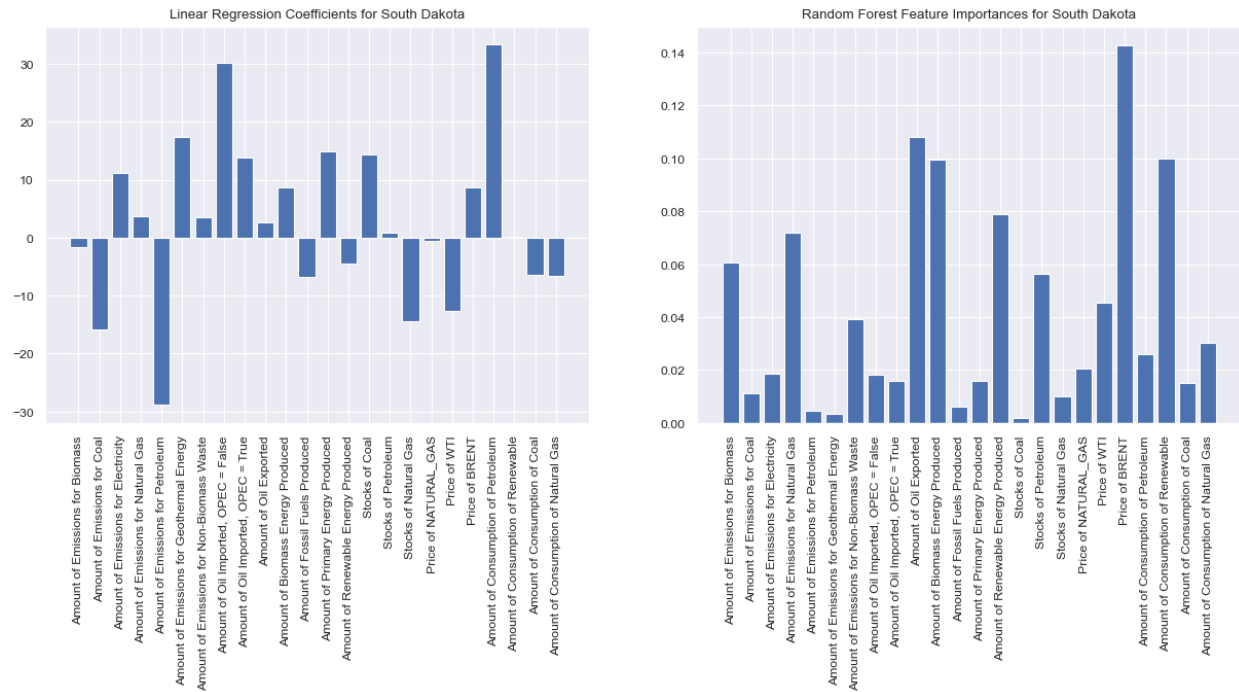


Figure 21: LASSO & Random Forest Regression Results for Physician & Clinician Expenditure in Dakota

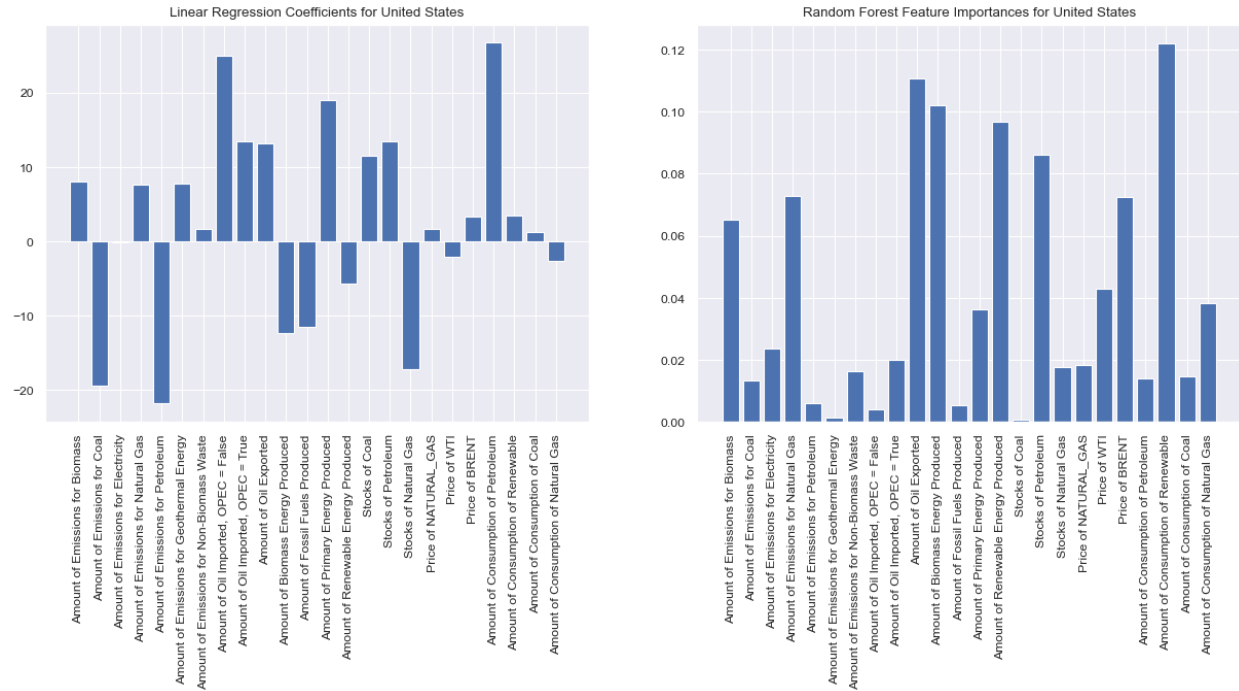


Figure 22: LASSO & Random Forest Regression Results for Physician & Clinician Expenditure in the US

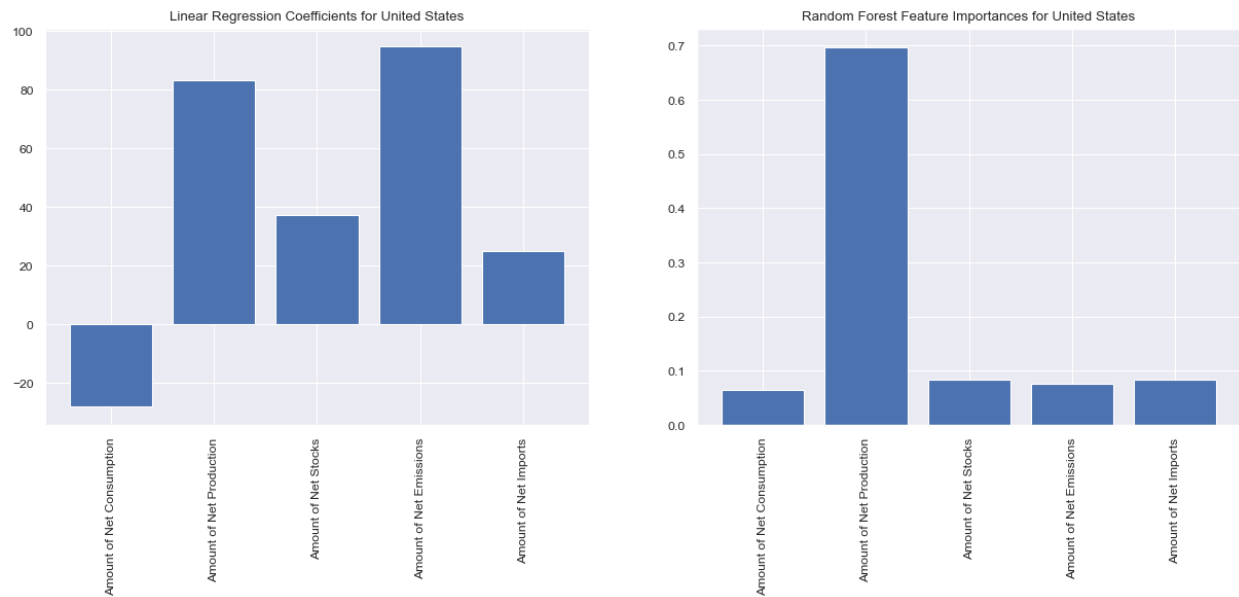


Figure 23: LASSO & Random Forest Regression Results for Physician & Clinician Expenditure in the US, using only aggregate predictors

4.2 Forecasting Oil Prices

We noticed that energy prices were moderately significant predictors in our models, especially when used to predict unemployment. Hence, as a potential extension of our report, we attempted to create a model for predicting crude oil prices.

4.2.1 Method

We decided to accomplish this task using a Long Short-Term Memory (LSTM) neural network model. In particular, we use feature engineering to create a predictor for the average daily crude oil price (in USD) based on the daily spot prices of Brent crude oil (Brent) and West Texas Intermediate crude oil (WTI) prices between January 2nd 1986 and January 22nd 2024. Note that we dropped rows for which the markets were closed.

To train our model, we pre-processed the data by scaling the average crude oil price values to a $[0, 1]$ range using the `MinMaxScaler`. This normalization is important for LSTM models to perform optimally. Next, we created a dataset with a time lag of three days (`look_back=3`) so that the model could learn to predict the price of oil on the next day based on the prices of the lagged datapoints. We then split this dataset into training (80%) and testing (20%) sets to validate the model's performance. In other words we trained our model on the average daily crude oil prices between January 2nd 1986 and July 26th 2016. For the LSTM model architecture, we used two LSTM layers, each followed by a dropout layer to prevent over-fitting. The model concluded with a dense layer to output the predicted price. We compiled the model with the Adam optimizer and used mean squared error as our loss function. We trained for 50 epochs with a batch size of 32. With the limited time scope of this project, we did a non-exhaustive search of these hyper-parameters and found the given configuration to work well, but acknowledge that in future work the performance can be further improved.

4.2.2 Results

Our model achieves a R-squared value of 0.89, a Train-RMSE of 7.062 and a Test-RMSE of 6.2. We used RMSE for evaluation to transform the errors back into the original unit (USD) for interpretability. The predictions for our model are displayed in Figure 24 and 25. We can see that the predictions of our model track the actual prices reasonably well with the exception of two outlier events: First, during the COVID-19 pandemic in 2020 (Monday, April 21, 2020) the oil prices momentarily fell below 0 for the first time in history. Our model predicted a value close to 0 for this event. Second, in 2022 the oil prices rallied after the invasion of Ukraine by Russia due to geopolitical concerns, including the risk of an embargo of Russian crude oil.

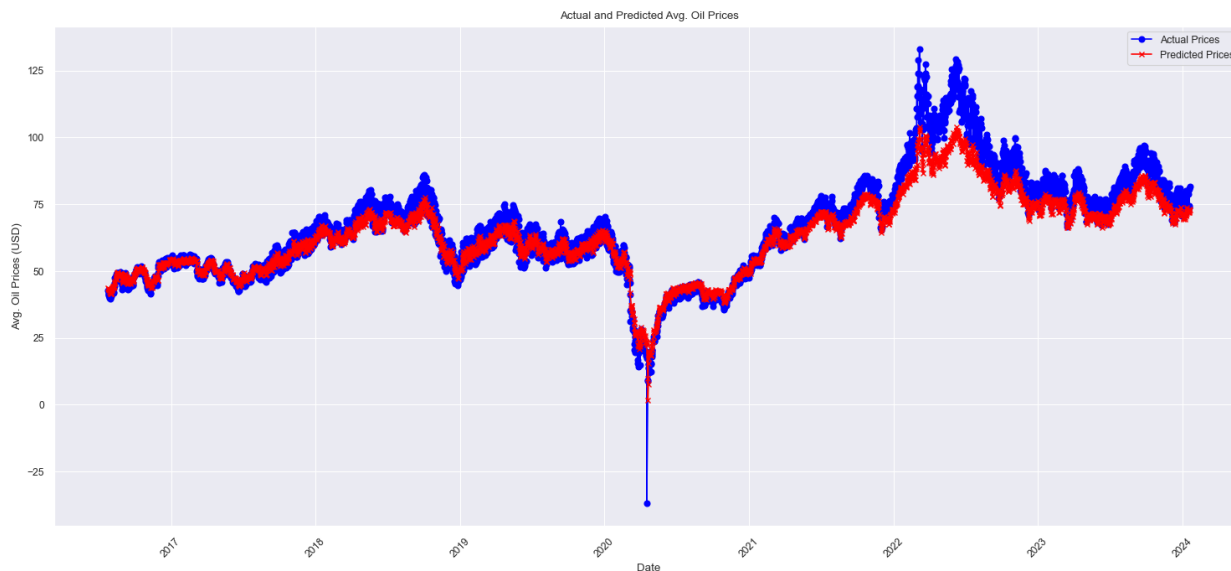


Figure 24: Predicted vs. actual average daily crude oil prices (Brent and WTI) in USD between 1986-07-26 and 2024-01-19

4.2.3 Discussion

Our results suggest that the LSTM model is able to learn the aggregate trend of crude oil prices, achieving a high R-squared value a relatively low RMSE. This suggests strong predictive performance for the model. However, the limitations in predicting extreme market events, like the negative prices during the COVID-19 pandemic or the sharp increase following geopolitical tensions, highlight the need for future improvements. In particular, we hypothesize that incorporating additional features such as economic indicators, production levels, geopolitical events, and market sentiment could enhance the model’s ability to predict these outliers.

This exploration demonstrates the potential of LSTM models in predicting commodity prices and sets the groundwork for more sophisticated predictive models in the future, which could be used to forecast unemployment and other socioeconomic indicators, allowing policymakers to prepare ahead as the US transitions towards a green economy.

5 Conclusion

The United States stands at a critical juncture in its energy policy, balancing the economic benefits of fossil fuel consumption and production against the urgent need for a transition to renewable energy.

The significant influence of petroleum consumption and oil exports on unemployment and income highlights the unfortunate reality of the US’ economic reliance on fossil fuels. Conversely, the positive correlation between renewable energy consumption and production with employment and income in certain greener states signals a promising shift towards sustainable economic growth.

We posit two further areas of investigation: Firstly, since we did not yield definitive conclusions on health-care expenditure, but managed to identify energy production in general as an important predictor, further work can be done on how exactly energy production impacts health in the united states. Additionally, our proof-of-concept prediction of crude oil prices through an LSTM model demonstrates how ML techniques could be used to predict and safeguard against the negative side effects of transitioning to a green economy.

6 Appendix A: List of Energy-Related Variables

Aggregate energy-related variable	Sub-variables
Energy consumption	Amount of consumption of petroleum Amount of consumption of renewables Amount of consumption of coal Amount of consumption of natural gas
Energy production	Amount of biomass energy produced Amount of fossil fuels produced Amount of primary energy produced Amount of renewable energy produced
Net energy stocks on hand	Stocks of coal Stocks of petroleum Stocks of natural gas
Net emissions	Amount of emissions for biomass Amount of emissions for coal Amount of emissions for electricity Amount of emissions for natural gas Amount of emissions for petroleum Amount of emissions for geothermal energy Amount of emissions for non-biomass waste
Net oil imports	Amount of oil imported from non-OPEC nations Amount of oil imported from OPEC nations Amount of oil exported
Energy prices	Price of natural gas Price of WTI Price of Brent

7 Appendix B: LSTM Model Performance

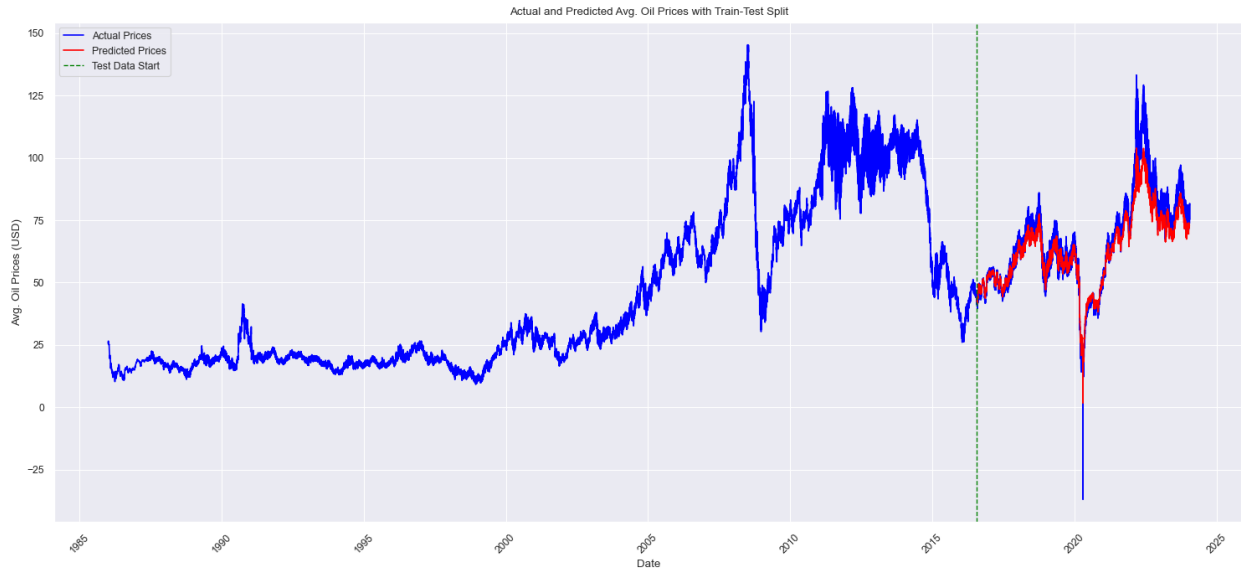


Figure 25: Predicted vs. actual average daily crude oil prices (Brent and WTI) in USD between 2016-01-02 and 2024-01-19. The dashed green line indicates the start of the test data.

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